

Equal Accuracy, Unequal Cost: Channel Dependence in PatchTST under Controlled Coupling

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Abstract

Channel-independent (CI) and channel-dependent (CD) variants of PatchTST are typically compared on observational benchmarks where correlation strength, dimensionality, and temporal dynamics all vary at once, leaving no clean way to attribute either variant’s win to any single cause. We isolate one factor at a time instead. A factorial AR(1) experiment varies contemporaneous correlation ρ and variate count C while Granger non-causality holds across channels by construction; a leader-follower VAR(1) experiment then introduces lag-1 coupling of controllable strength γ . Across both experiments the result is simple: letting the model attend across channels buys no measurable accuracy advantage on these controlled generative processes, at a substantial increase in training cost. On the synthetic grid we detect no consistent CD advantage at the 1% level: the grand-mean CD–CI difference is +0.0013 MSE (seed-clustered 95% CI $[-0.0023, +0.0048]$) over five seeds, with a pooled within-cell detection half-width of 0.0063 MSE. Any global CD gain (averaged across the nine AR(1) cells) above roughly 0.62% of the CI mean would have moved the interval off zero, and none did. Under the original early-stopping protocol an apparent CD deficit emerges under lagged coupling and grows with γ , staying inside our 1% band throughout. An instrumented diagnostic at the leader-follower $\gamma = 0.6$ cell documents a mechanism without explaining the deficit: CD reaches a sharper, earlier validation minimum than CI and degrades past it, but the protocol already restores validation-best weights, so the gap cannot be a selection artefact. Fresh reruns of that cell give CD–CI of -0.0019 MSE (95% CI $[-0.0105, +0.0066]$) against the original +0.0079, a run-to-run spread comparable to the effect itself at five seeds. A closed-form analysis of the generative model shows this near-parity is analytically expected, not merely observed: at the evaluation horizon, no predictor, CI or CD, retains any cross-channel signal to exploit. A cross-variate prediction head ties CI at every γ and removes the coupling-dependent widening, consistent with that widening originating in the shared head rather than the encoder. Matching the gradient-update budget across modes also removes the deficit, arguing against the per-epoch step-count asymmetry as its source, and a block-covariance family extends the same no-advantage finding beyond compound-symmetry correlation. Across every tested synthetic cell (AR(1), leader-follower, and block-covariance) the absolute CD–CI difference stays inside a pre-specified 1% relative-MSE band in both directions. Cost is another matter entirely: CD requires smaller feasible batches and far more gradient steps per epoch, its batch size collapses to one at $C = 84$ in the main grid, and at $C = 321$ on ECL (Electricity Consuming Load) a CD epoch costs roughly 956s at batch 8; in a single-seed comparison at $H \in \{96, 336\}$, CD trails CI by 5.6–7.6%. On the observational ETTh1 dataset (hourly electricity-transformer readings, $C = 7$), a matched-budget comparison still gives CI a modest but reliable edge (pooled CD–CI of +0.0147 MSE, seed-clustered 95% CI $[+0.0000, +0.0293]$, significant at horizons 96 and 720). CI comes out as the better default on accuracy per unit compute. All findings are scoped to PatchTST’s flattened-token CD formulation, the generative families studied, and a single-T4 training budget.

1 Introduction

Whether channel-independent (CI) or channel-dependent (CD) architectures forecast better has been judged almost entirely on observational benchmarks, where many properties of the data move together across datasets. Such a win identifies which model is stronger on those particular series, but not why. Here we control the generative process instead: synthetic AR(1) and leader-follower VAR(1) families that vary instantaneous correlation, lagged cross-channel dependence, and variate count independently, with ETTh1 and ECL serving as real-data context rather than primary evidence.¹

Section 2 describes the experimental design, Section 3 presents the AR(1) grid and real-data results, Section 4 covers the leader-follower experiments, Section 5 presents the fair-protocol robustness controls, Section 6 derives the closed-form accuracy ceilings, and the remaining sections cover limitations, related work, and conclusions.

1.1 Research Questions and Scope

CI architectures like PatchTST have become strong baselines for multivariate time series forecasting despite ignoring cross-channel interactions in their attention layers (Nie et al., 2023). At the same time, iTransformer, LIFT, and Abdelmalak et al. each report settings where modelling channel dependence explicitly yields clear gains (Liu et al., 2024; Zhao & Shen, 2024; Abdelmalak et al., 2025). None of these results are directly comparable, since the architectures, datasets, and notions of cross-channel structure differ across papers, leaving no account of which property of the data tips the balance.

We focus on one specific part of that question: multivariate processes in which channels share *instantaneous* correlation but have no lagged cross-channel dependence in the dynamics, using the AR(1) design introduced above, where ρ controls contemporaneous correlation while Granger non-causality holds across all channel pairs by construction. Two questions frame the work:

RQ1. In a setting where channels share purely instantaneous correlation, even at high ρ and high dimensionality, and no lagged cross-channel dependence exists by construction, does a channel-dependent PatchTST variant achieve meaningfully better forecast accuracy than its channel-independent counterpart?

RQ2. When lagged cross-channel coupling is introduced at controlled strength γ via a leader-follower VAR(1) process, does CD outperform CI consistently as γ grows, and does this depend on the patch resolution used?

RQ1 is answered by a $C \times \rho \times \text{mode}$ factorial experiment on AR(1) data where Granger non-causality holds by construction. RQ2 is answered by two companion experiments: a γ sweep at fixed patch size $P = 16$, and a joint sweep over $\gamma \in \{0.0, 0.3, 0.6, 0.9\}$ and $P \in \{2, 4, 8, 16\}$ testing whether finer patch resolution concentrates lag-1 coupling within individual patches and thereby gives CD’s attention a detectable signal. That second sweep does not cover its full grid: CD trains at $P \in \{4, 8, 16\}$, but at $P = 2$ the flattened CD sequence reaches $21 \times 511 = 10,731$ tokens and exceeds T4 memory at any batch size under the non-fused attention implementation used for this paper’s runs, so the finest resolution (the one where a lag-1 relation falls across adjacent patches) carries CI runs only. RQ2 is therefore answered over $P \in \{4, 8, 16\}$ and left open at $P = 2$.

Within the regimes actually tested, both RQ1 and RQ2 come back null at our 1% practical threshold: small differences exist, but none clears the bar. Section 6 establishes that this null is what the generative model predicts: at the evaluation horizon $h = 96$, the analytical gap between a full-information and a channel-restricted predictor has closed to zero width for every tested γ , leaving no predictor of either kind with cross-channel signal to exploit. The comparison at $h = 96$ therefore cannot discriminate the architectures even in principle, and RQ2’s null is scoped to that horizon; the short-horizon regime, where the ceilings

¹Code, configuration files, result files, and analysis scripts for reproducing all reported results are available at <https://github.com/AdiMendelowitz/ci-cd-patchtst> (release v1.0-tmlr).

separate, is left open and named in the Limitations. A set of fair-protocol controls (Section 5) then shows that the apparent CD deficit under lagged coupling does not survive a matched update budget or a cross-variate head, and is small relative to run-to-run variation at five seeds.

Purely instantaneous correlation, regardless of strength or variate count, is not sufficient for CD to gain an advantage over CI under PatchTST’s architecture, and this null extends to lag-1 coupling at the strengths tested and at patch resolutions $P \in \{4, 8, 16\}$ (the $P = 2$ CD configuration was infeasible under the non-fused attention implementation used for this paper’s runs).

2 Experimental Setup

2.1 Synthetic Data

We generate multivariate time series from a vector AR(1) process:

$$x_t = \phi x_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \Sigma), \quad (1)$$

where $\phi = 0.8$ and Σ has compound-symmetry structure: $\Sigma_{ii} = 1$ and $\Sigma_{ij} = \rho$ for $i \neq j$. The parameter ρ directly controls pairwise contemporaneous correlation while the transition matrix stays diagonal. We discard 1000 burn-in steps before recording observations.

For each configuration in this grid we generate 14,400 timesteps (13,400 for seeds {42, 123, 456}; Table 1), split 60/20/20 into train, validation, and test sets; the table also gives the corresponding figures for the other synthetic families. Every split is normalised with per-channel z-score statistics fit on the training set only. Three factors vary independently: number of channels $C \in \{7, 21, 84\}$, correlation strength $\rho \in \{0.1, 0.5, 0.9\}$, and model mode $\in \{\text{CI}, \text{CD}, \text{DLinear}\}$, with five random seeds {42, 123, 456, 789, 1011} per cell, for 135 runs in total. We report the empirical pairwise Pearson correlation of the training data alongside the nominal ρ to confirm the data-generating process (DGP) behaves as intended.

The four synthetic families in this paper do not all share one generator. The AR(1) grid, the block-covariance family, and the leader-follower γ -sweep use the process specified above, differing only in the innovation covariance, the transition matrix, and (within the AR(1) grid) the series length, which takes two values across its five seeds. The (P, γ) boundary sweep of Section 4 uses a separate generator with iid innovations, a longer series, and a different split. Table 1 states each family’s protocol explicitly. Because the boundary protocol differs from the other three simultaneously, absolute MSE levels cannot be compared between it and the other three families, and we make no such comparison; within each family the CD/CI ratio is computed against a CI arm trained under that same protocol, keeping the ratio comparable throughout.

Dynamical structure. The transition operator $\phi\mathbf{I}$ is diagonal, so $A^k = (\phi\mathbf{I})^k = \phi^k\mathbf{I}$ has zero off-diagonal entries at every power: each channel’s future depends only on its own past, and Granger non-causality holds exactly across all channel pairs. Cross-variate dependence, where present, comes entirely from the contemporaneous covariance Σ , not from the dynamics – and even that dependence carries no Granger-predictive information, since Σ correlates innovations that are unobserved at prediction time. Lagged cross-covariance between Σ -correlated channels need not vanish at every k , but nothing in it is exploitable beyond each channel’s own history. The leader-follower experiments in Section 4 introduce the complementary case, where the transition operator structurally links one variate’s future to another’s past.

Covariance design. Compound-symmetry assigns the same pairwise correlation ρ to every channel pair, giving CD no differential block structure or correlation hierarchy to exploit. This is the configuration most hostile to CD under instantaneous-only coupling: if CD cannot gain an advantage here, contemporaneous correlation alone will not suffice under any other covariance structure within this family.

Prediction horizon. All synthetic AR(1) runs use `pred_len = 96`, matching the shortest horizon in the ETTh1 experiments.

Table 1: Data-generating protocols for the four synthetic families. All are VAR(1) at $\phi = 0.8$, use $L = 512$ and $H = 96$, and normalise by per-channel z-score statistics fit on the training split only. Three families share one protocol, differing only in covariance and transition structure; the boundary sweep differs further, though not in innovation scale ($\sigma = 0.1$ versus unit is removed by z-scoring) or the absence of burn-in (immaterial given $\phi = 0.8$'s fast mixing over 20,000 steps). What is substantive is the innovation covariance, series length, and split. The boundary family's recorded $\rho = 0.5$ is a schema constant kept for mergeability, not a data property: its innovations are iid, with no contemporaneous cross-channel correlation.

	AR(1) grid (Sec. 2)	Block-cov. (Sec. 5)	Leader-follower γ -sweep (Sec. 4)	Boundary (P, γ) (Sec. 4)
Transition matrix	$\phi \mathbf{I}$	$\phi \mathbf{I}$	$A(\gamma)$ lower-tri	$A(\gamma)$ lower-tri
Innovation cov.	comp. sym. ρ	block $\rho_{\text{in}}, 0$	comp. sym. $\rho = 0.5$	iid (diagonal)
Innovation scale	unit	unit	unit	$\sigma = 0.1$
Burn-in steps	1000	1000	1000	none
Timesteps T	13,400 / 14,400 ^c	14,400	14,400	20,000
Split	60/20/20 %	60/20/20 %	60/20/20 %	70/10/20 %
Train windows ^a	7,433 / 8,033 ^c	8,033	8,033	13,393
Partial final batch	kept	kept	kept	dropped
Channels C	{7, 21, 84}	{21, 84}	21	21
Coupling γ	—	—	{0, .3, .6, .9}	{0, .3, .6, .9}
Patch P / stride S	16 / 8	16 / 8	16 / 8	$P / P/2, P \in \{2, 4, 8, 16\}$
Modes ^b	CI, CD, DLinear	CI, CD, DLinear	CI, CD, DLinear	CI, CD

^a Training windows = $\lfloor T \cdot f_{\text{train}} \rfloor - L - H + 1$ at $L = 512, H = 96$.

^b Modes attempted. Which cells completed under compute constraints is stated per experiment.

^c AR(1) grid only: seeds {42, 123, 456} were generated with 13,400 usable timesteps and seeds {789, 1011} with 14,400, giving 7,433 and 8,033 training windows respectively. Per-seed CD–CI differences are unaffected, since both modes of a given seed share that seed's series; grid means in Table 4 average over both. Table 3 reports the 7,433 figures.

2.2 Real-World Datasets

ETTh1. Hourly measurements of 7 variates (oil temperature and six power-load features) from a Chinese electrical transformer. We use the standard train/val/test split of 8640/2880/2880 timesteps. The mean absolute off-diagonal Pearson correlation at lag 0 across the training split is 0.311 (median 0.233), indicating moderate contemporaneous cross-variate correlation.

ECL (Electricity Consuming Load). Hourly electricity consumption for 321 clients over 26,304 timesteps. (The iTransformer paper (Liu et al., 2024) reports 26,352 rows; the difference reflects a minor dataset version discrepancy and does not affect the experimental protocol.) We apply proportional 60/20/20 splitting following Liu et al., yielding 15,813 train / 5,247 validation / 5,244 test rows. The mean absolute off-diagonal Pearson correlation at lag 0 is 0.503 (median 0.483), substantially higher than ETTh1. All data are normalised with per-channel z-score statistics fit on the training set only.

2.3 Models

PatchTST is the backbone for both CI and CD modes because the two modes differ only in whether self-attention operates within each variate's patch sequence independently or across all variates' patch sequences jointly. Patch extraction, linear projection, Transformer encoder, training procedure, and hyperparameters are otherwise shared, making this the most direct available test of the CD/CI distinction without entangling it with other architectural choices. iTransformer would confound the comparison with its inverted embedding design, and Crossformer with its two-stage routing attention. Every finding here is conditioned on PatchTST's patch-based representation and the lookback window and patch granularity used.

We implement PatchTST (Nie et al., 2023) from scratch in PyTorch, supporting both modes.

Channel-Independent (CI). Each variate is processed separately. An input of shape (B, L, C) is reshaped to $(B \cdot C, L)$, split into patches of length P with stride S , and passed through a shared Transformer encoder, which sees N patches per variate.

Channel-Dependent (CD). All variates are processed jointly. After patching, tokens from every variate are concatenated along the sequence dimension before the encoder, giving an input of shape $(B, C \cdot N, D)$ over which self-attention operates simultaneously, letting the model attend across variates. The encoder output is then split back into C per-variate representations, and a shared linear head projects each to the prediction horizon independently, so only the encoder mixes cross-variate information. Every CD result reported in this paper uses tokens that carry no explicit channel-identity or temporal-position embedding – both come from fixed ordering alone, channel-major for CD (all N patches of channel 0, then channel 1, and so on) and the natural patch sequence for CI.

For the synthetic experiments and ETTh1 we use $L = 512$, $P = 16$, $S = 8$, giving

$$N = \left\lfloor \frac{L - P}{S} \right\rfloor + 1 = \left\lfloor \frac{512 - 16}{8} \right\rfloor + 1 = 63 \quad (2)$$

patches per variate. For ECL we use $L = 96$ following iTransformer (Liu et al., 2024), giving $N = 11$ patches per variate and $C \cdot N = 321 \times 11 = 3531$ tokens for CD. All remaining architectural hyperparameters are listed in Table 2.

Table 2: Architectural hyperparameters. Synthetic and ETTh1 share one configuration; ECL uses a wider model to accommodate 321 variates.

Hyperparameter	Synthetic + ETTh1	ECL
d_{model}	64	128
Attention heads	8	16
Encoder layers	3	3
Dropout	0.2	0.2
Sequence length L	512	96
Patch size P	16	16
Stride S	8	8
Patches per variate N	63	11

We include DLinear (Zeng et al., 2023) as a linear baseline. DLinear decomposes each variate’s lookback window into trend (moving-average filter with kernel size 25) and remainder, then applies separate channel-independent linear maps from L to H for each component, summing the two outputs. It is the most competitive non-Transformer baseline on standard benchmarks and sets a natural lower bar for what qualifies as learned temporal structure. The DLinear values in Tables 4 and 7 come from a dedicated re-run using this full decomposition; CI and CD values in Table 7 come from a separate set of runs using the same windowing and seeds.

2.4 Training Regime

PatchTST (CI and CD) is trained with AdamW (lr = 10^{-4} , weight_decay = 10^{-4}), a cosine learning-rate schedule with linear warm-up, gradient clipping at max_norm = 1.0, and FP16 mixed precision (autocast + GradScaler). DLinear uses the same optimiser without warm-up. Early stopping is applied on validation MSE with patience 10 and a maximum of 100 epochs for ETTh1 and 50 epochs for all other experiments; the validation-best checkpoint is restored and evaluated in every case. All experiments run on a single NVIDIA T4 16 GB GPU; Kaggle’s dual-T4 accelerator option was selected for every session, but no run uses more than one GPU through model or data parallelism, so every reported wall-clock and memory figure reflects one T4’s actual utilisation.

Batch sizes. T4 memory constraints force different batch sizes between CI and CD at large C , creating an asymmetry in per-epoch gradient update counts. Table 3 reports these counts for each configuration.

Table 3: Batch sizes and gradient update counts per epoch by mode and C . The Ratio column is CD steps / CI steps. At $C = 84$, steps/epoch reflects the 7,433-window figure (Table 1, footnote c); CD/CI ratios are unaffected by the seed-dependent window count, since both modes of a given seed share it.

C	CI batch	CD batch	CI steps/epoch	CD steps/epoch	Ratio
7	128	64	59	117	$2.0\times$
21	128	8	59	930	$15.8\times$
84	32	1	233	7433	$31.9\times$

For ETTh1 both modes use batch size 32 under the matched-budget protocol below. For ECL both modes use batch size 8 (1,953 steps per epoch at $H = 96$; 1,923 at $H = 336$).

Step-count confound. At $C = 84$, CD runs with batch size 1, giving $31.9\times$ more gradient updates per epoch than CI. The cause is a hardware memory limit, not a deliberate experimental choice. We log `total_steps` for every run and report it alongside results; Section 3 discusses what this asymmetry means for the null result.

Matched-budget ETTh1 protocol. Both CI and CD use batch size 32, `warmup_epochs` = 10, and a minimum of 15 epochs before early stopping may fire. The identical batch size gives both modes the same number of gradient updates per epoch at every horizon (252, 249, 244, and 232 at $H \in \{96, 192, 336, 720\}$ respectively), removing the step-count confound present in earlier runs. The minimum-epoch floor prevents a degenerate epoch-one stop without inflating test MSE, since the best validation checkpoint is still used for evaluation.

ECL CD cost. At $C = 321$ and $L = 96$ the flattened CD sequence is 3,531 tokens, and a CD training epoch takes roughly 956s on the T4 at batch 8 (measured at $H = 336$), against roughly 143s for a CI epoch at the same horizon and batch size, a $6.7\times$ cost ratio; a worst-case 50-epoch run would exceed a single 12-hour session on wall-clock time. Early stopping keeps this well within budget in practice – the completed CD runs stopped at epochs 23 ($H = 96$) and 17 ($H = 336$), the latter costing about 4.5 GPU-hours – and per-epoch checkpointing with cross-session resumption covers the worst case regardless. Throughout this paper, *commodity GPU budget* refers to single-T4 training with a 12-hour wall-clock limit, as in a standard Kaggle notebook session (see the single-device disclosure in Section 2).

VRAM cost of the memory wall. The token-count argument above can be made concrete. A naive (non-fused) attention implementation materialises the full $(C \cdot N) \times (C \cdot N)$ score matrix per head per layer; at ECL’s architecture ($C \cdot N = 3,531$ tokens), this term alone requires 47.6 GiB for one encoder layer at batch 128 and 142.7 GiB across all three layers if retained simultaneously, both well beyond a T4’s 16 GiB budget. Measured peak allocation is 8.13 GiB at batch 128 and 0.53 GiB at batch 8, roughly $17\times$ and $16.7\times$ below the corresponding three-layer bounds of 142.7 and 8.9 GiB. Fused attention kernels in current PyTorch never materialise this term, instead computing attention with $O(C \cdot N)$ working memory per head via online softmax, which accounts for the entire gap. The compute cost, $O(C^2 N^2)$, is unaffected by kernel fusion and remains the binding constraint at this scale; the finer-patch configurations examined in Section 4 hit the same memory wall this bound describes, under the non-fused implementation used for this paper’s runs.

3 Results

3.1 Synthetic AR(1) Grid

Table 4 shows mean test MSE over five seeds for each cell. CI and CD track each other almost exactly across the grid, with the largest mean difference of 0.0054 MSE units at $C = 84$, $\rho = 0.9$, and no consistent direction to the gap. CI leads at $C = 7$ across all ρ , at $C = 21$ for $\rho \in \{0.1, 0.5\}$, and at $C = 84$ for $\rho \in \{0.5, 0.9\}$; CD leads marginally at $C = 21$, $\rho = 0.9$ and at $C = 84$, $\rho = 0.1$. At $C \in \{21, 84\}$ DLinear trails both PatchTST variants everywhere except at the $C = 84$, $\rho = 0.1$ cell, where all three modes coincide at 0.9831–0.9832. At $C = 7$ it trails both variants throughout, by roughly 1% at $\rho = 0.1$, widening to roughly 4% at $\rho = 0.9$. The

Transformer representations add value as dimensionality and correlation strength rise, but this leaves the CD/CI comparison (the central question) unaffected by where the DLinear baseline falls.

Table 4: Mean test MSE over seeds {42, 123, 456, 789, 1011} for the synthetic AR(1) grid. CD/CI above 1.0 favours CI; below 1.0 favours CD.

C	ρ	CI	CD	DLinear	CD/CI
7	0.1	1.0282	1.0298	1.0404	1.0016
7	0.5	1.0404	1.0412	1.0557	1.0008
7	0.9	1.0642	1.0653	1.1064	1.0011
21	0.1	0.9931	0.9950	1.0013	1.0019
21	0.5	1.0179	1.0189	1.0339	1.0010
21	0.9	1.0327	1.0311	1.0595	0.9984
84	0.1	0.9832	0.9831	0.9831	0.9999
84	0.5	0.9752	0.9764	0.9781	1.0012
84	0.9	0.9759	0.9813	0.9942	1.0056

Synthetic AR(1) grid: CD/CI MSE ratio
(>1.0 favours CI, <1.0 favours CD; colour scale fixed at $\pm 0.5\%$)

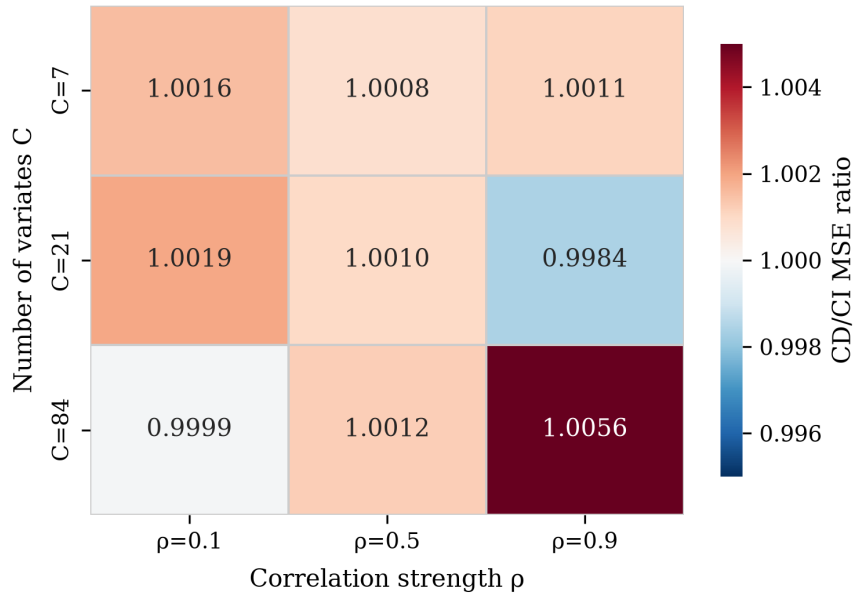


Figure 1: Synthetic AR(1) grid: CD/CI test-MSE ratio by variate count C and contemporaneous correlation ρ , averaged over seeds {42, 123, 456, 789, 1011}. Values above 1.0 favour CI, below 1.0 favour CD; the colour scale is fixed at $\pm 0.5\%$. No cell departs from unity by more than 0.56%, and the sign of the deviation is not consistent across the grid.

The $C = 84$, $\rho = 0.9$ cell deserves closer attention. Seeds 42, 123, and 456 all stopped at best epoch 1 or 2 (7433 / 14866 / 7433 total gradient updates), while CI stopped with only 233–466 updates; at $n = 5$, CD’s mean is 0.9813 against CI’s 0.9759. The same pattern holds at the other two $C = 84$ cells, where every CD seed stops earlier than every CI seed in the same cell (CD best epochs 4–10 against CI’s 21–28 at $\rho = 0.1$; 4 against 9–12 at $\rho = 0.5$). Across all three cells, CD consumes up to $32\times$ more compute per epoch and more epochs before stopping, yet the two modes converge to essentially the same test MSE. The extra gradient updates do not help CD, because there is no lagged cross-channel signal for its attention to exploit.

Statistical analysis. For each (C, ρ) cell we compute the per-seed difference $d_s = e_s^{\text{CD}} - e_s^{\text{CI}}$ over seeds $\{42, 123, 456, 789, 1011\}$ and derive its mean, standard deviation, and 95% confidence interval from a t -distribution with four degrees of freedom ($t_{0.025,4} = 2.776$). Across all nine cells the mean CD–CI difference stays within ± 0.0054 MSE units, at most 0.56% of the corresponding CI mean, and seven of nine 95% CIs include zero; the two that exclude zero, at $C = 7, \rho = 0.1$ and $C = 21, \rho = 0.1$, are both positive (CD worse) and below 0.19% of the CI mean. Pooling the 45 matched pairs as a simple unweighted average gives a grand-mean CD–CI of $+0.0013$ MSE (0.13% of the mean CI MSE), and regressing the paired difference on C (categorical) and ρ finds no dependence on either factor (F -test $p = 0.81, R^2 = 0.02$). Because the same five seeds recur across all nine cells, we report the grand mean’s interval clustered by seed rather than treating all 45 rows as independent – averaging each seed’s difference across the nine cells first gives a wider but still zero-including 95% CI of $[-0.0023, +0.0048]$. Pooling the within-cell variances gives a 95% CI half-width of 0.0063 MSE at $n = 5$, roughly 0.62% of the mean CI MSE. Any uniform CD advantage larger than that would have pushed the grand-mean CI entirely above zero, and none of the observed differences come close, including the two per-cell exceptions above.

3.2 Real-Data Experiments

ETTh1 ($C = 7$). Under the matched-budget protocol, CI matches or beats CD at every horizon (Table 5). CD is significantly worse at $H = 96$ (paired 95% CI $[+0.0013, +0.0618]$) and $H = 720$ (95% CI $[+0.0041, +0.0367]$); the $H = 192$ and $H = 336$ differences are not significant. Pooled across all 20 pairs (four horizons, five seeds), the CD–CI difference is $+0.0147$ MSE. The same five seeds recur at every horizon, so we report this pooled figure’s interval clustered by seed rather than treating all 20 pairs as independent draws – averaging each seed’s difference across the four horizons first gives a 95% CI of $[+0.0000, +0.0293]$. At $H = 192$ the ratio sits below one (CD 0.5194 vs. CI 0.5229), the only horizon where CD leads, but the 95% CI $[-0.0227, +0.0159]$ includes zero, meaning CD’s lead there is not reliable. Steps per epoch are identical across modes at every horizon, so neither the update count nor the warmup schedule can explain the CI advantage at the significant horizons.

Table 5: ETTh1 mean test MSE over seeds $\{42, 123, 456, 789, 1011\}$, matched-budget protocol (batch 32, warmup 10, min epochs 15 for both modes). CD/CI above 1.0 favours CI. SIG = paired 95% CI excludes zero.

H	CI	CD	CD/CI	
96	0.4599	0.4914	1.0686	SIG
192	0.5229	0.5194	0.9935	
336	0.5789	0.5890	1.0174	
720	0.7062	0.7266	1.0289	SIG

ETTh1 subgroups by lagged coupling strength. To test whether CI’s edge concentrates where lagged cross-channel coupling is strongest, we partition test windows by lagged coupling strength, using the mean absolute lag-1 cross-correlation of first differences as the primary metric, specified before the error data existed, with the raw lag-1 statistic serving as a robustness check since the two rank windows in near-opposite order on this data. The metric is strongly autocorrelated across window starts, so a median split forms contiguous time regimes rather than an interleaved sample, and any subgroup gap is therefore partially confounded with regime; inference aggregates errors to run level per subset before pairing by seed. Under the primary split, the CD–CI contrast is $H = 96$ high-coupling $+0.0058$ (95% CI $[-0.0159, +0.0276]$) against low-coupling $+0.0450$ ($[+0.0208, +0.0692]$); $H = 192$ high $+0.0183$ ($[-0.0151, +0.0517]$) against low $+0.0466$ ($[-0.0258, +0.1190]$); $H = 336$ high -0.0030 ($[-0.0259, +0.0199]$) against low $+0.0167$ ($[-0.0089, +0.0423]$); $H = 720$ high $+0.0263$ ($[+0.0196, +0.0329]$) against low $+0.0302$ ($[+0.0275, +0.0330]$), all $n = 5$ paired over seeds. The low-coupling subgroup’s point estimate matches or exceeds the high-coupling subgroup’s at every horizon, the opposite of what a coupling-driven-advantage account would predict, so we do not read CI’s ETTh1 edge as concentrated in windows of higher lagged cross-channel coupling. This rules out a coupling-exploitation account of CI’s edge, though the temporal confounding noted above means the underlying mechanism stays unidentified.

ECL ($C = 321$). Both modes train at this scale under the single-T4 budget (Section 2), and cost is what separates them. The accuracy comparison is deliberately narrow, one seed (456) at $H \in \{96, 336\}$, and is reported descriptively with no confidence intervals (Table 6). CD trails CI at both horizons, by +0.0133 MSE (+7.6% of the CI value) at $H = 96$ and +0.0113 (+5.6%) at $H = 336$, and reaches its validation minimum far earlier (epochs 13 and 7 against CI’s 42 and 25) – the same early-minimum shape documented at the synthetic $\gamma = 0.6$ cell (Section 5.1). ECL is included as evidence of this formulation’s cost asymmetry at scale rather than as a performance benchmark; a single seed supports no significance claim, and we make none.

Table 6: ECL test MSE, CI versus CD at seed 456, $H \in \{96, 336\}$. Single seed, descriptive only; no confidence intervals are reported. Rel is $CD - CI$ as a percentage of the CI value.

H	CI	CD	CD−CI	Rel
96	0.1753	0.1886	+0.0133	+7.6%
336	0.2037	0.2150	+0.0113	+5.6%

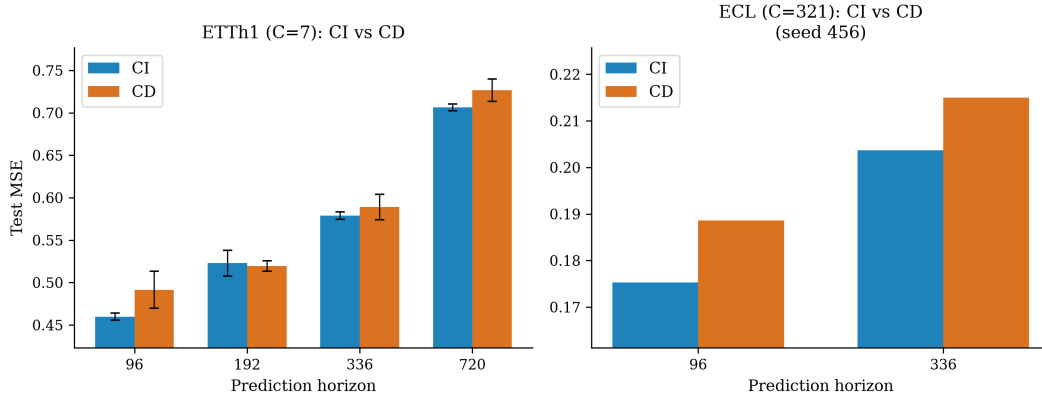


Figure 2: Real-data test MSE by prediction horizon. Left: ETTh1 ($C = 7$) across $H \in \{96, 192, 336, 720\}$, CI versus CD under the matched-budget protocol, mean over seeds $\{42, 123, 456, 789, 1011\}$ with error bars showing the cross-seed standard deviation; CI is significantly better at $H \in \{96, 720\}$ and never worse. Right: ECL ($C = 321$), CI versus CD at $H \in \{96, 336\}$, single seed (456); no error bars are drawn for single-seed groups.

Correlation and lead-lag structure. Beyond the lag-0 correlations reported in Section 2, lagged structure adds further signal in both datasets. For ETTh1, the mean of each pair’s maximum absolute correlation over lags is 0.349, and 43% of channel pairs gain at least 5 percentage points of absolute correlation from some non-zero lag; ECL’s figures run higher (mean max-over-lags 0.596, 49% of pairs gaining at least 5pp). In both datasets lag 0 remains the strongest single lag, but the lagged component is non-trivial for a substantial fraction of pairs.

Both datasets sit between the AR(1) grid (instantaneous correlation only) and the leader-follower regime (explicit structural lagged coupling). We treat the ETTh1 and ECL results as contextual evidence about CD/CI behaviour in realistic settings, not as controlled tests of the coupling-structure hypothesis.

4 Analysis: Leader-Follower Coupling Experiments

The AR(1) null result is easy to explain: when the data has no lagged cross-channel signal by construction, a model attending across channels has nothing additional to exploit beyond what channel-independent attention already sees. RQ2 asks whether structured lagged coupling changes that.

4.1 Leader-Follower VAR(1) Design

We generate data from a $C = 21$ VAR(1) process with the following channel structure: channels 0–9 are *leaders* (pure AR(1), no cross-channel dependence); channels 10–19 are *followers*, where follower $i + 10$ depends on leader i with lag-1 coefficient γ ; channel 20 is an *isolate* (pure AR(1), internal negative control). The transition matrix $A(\gamma)$ is lower-triangular with diagonal entries $\phi = 0.8$ and off-diagonal entries $A[i + 10, i] = \gamma$ for $i \in \{0, \dots, 9\}$. Lower-triangular structure guarantees spectral radius ϕ at all finite γ , so the process is stationary by construction.

At $\gamma = 0$ the process reduces to the original AR(1) grid at $C = 21$, $\rho = 0.5$, giving a cross-experiment sanity check. This anchor applies to the γ -sweep only; the (P, γ) boundary sweep uses a different generator (Table 1), so its $\gamma = 0$ cells are not expected to reproduce the AR(1) grid, and a mismatch there should not be read as a failed check. We sweep $\gamma \in \{0.0, 0.3, 0.6, 0.9\}$ with five seeds and CI, CD, and DLinear modes, all at patch size $P = 16$.

4.2 Patch-Resolution Sweep

The null result at $P = 16$ has a natural mechanistic explanation. With stride $S = 8$, each patch spans 16 timesteps, and a lag-1 cross-channel relationship (follower $i + 10$ at time t depending on leader i at time $t - 1$) means both timesteps likely fall within the same patch; whether the cross-variate attention can resolve a single-step offset within a 16-timestep window is unclear. Smaller patches would spread the lag-1 pair across adjacent patch tokens instead of collapsing it inside one, giving CD’s attention a more concentrated signal to work with.

We therefore run a joint sweep over $\gamma \in \{0.0, 0.3, 0.6, 0.9\}$ and $P \in \{2, 4, 8, 16\}$ with $S = P/2$ (50% overlap), modes CI and CD, five seeds. At $P = 2$, each patch spans exactly two timesteps and stride $S = 1$, so a lag-1 relationship falls across adjacent patches – the finest resolution the architecture supports. CD at $P = 4$ was not run in earlier tranches under an assumed VRAM ceiling at this sequence length ($C \cdot N = 21 \times 255 = 5355$ tokens); a dedicated boundary tranche, launched to test that assumption directly, trains CD successfully at $P = 4$ under the same single-device T4 budget, with full five-seed coverage now complete across all four γ (Table 8).

4.3 Results

Table 7 reports mean CI, CD, and DLinear MSE at $P = 16$ across the γ sweep ($C = 21$, $\text{pred_len} = 96$, five seeds). The CD/CI ratio stays within 1% of unity throughout and never falls below it. It does climb with γ under this early-stopping protocol, and regressing the paired CD–CI difference on γ gives a slope of $+0.0072$ MSE per unit γ (95% CI $[+0.0027, +0.0117]$, $p = 0.004$, $R^2 = 0.38$), though the climb never leaves the 1% band. Section 5.1 examines the $\gamma = 0.6$ cell directly and finds the gap is not produced by checkpoint selection, and is small relative to run-to-run variation at five seeds. DLinear stays within 0.004 MSE of CI at every γ , so at this C a channel-independent linear map is about as accurate as patch attention, though Table 4 shows the parity breaks at larger C or ρ . These DLinear runs reached the epoch budget without early stopping and should be read as budget-limited reference points rather than fully converged optima.

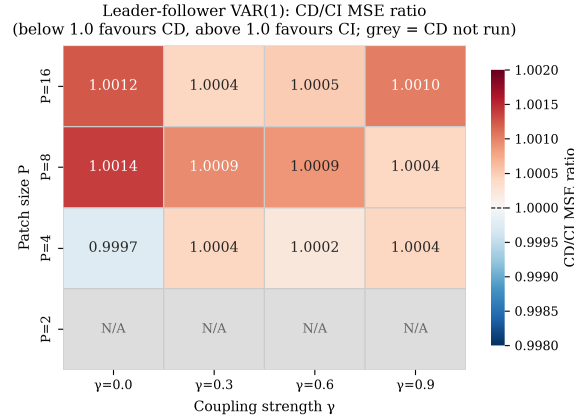
Table 7: Leader-follower VAR(1) at $P = 16$. Mean test MSE and CD/CI ratio over seeds $\{42, 123, 456, 789, 1011\}$. CI and CD match the AR(1) grid at $\gamma = 0.0$ to within seed variance. The CD/CI ratio stays within 1% of unity across γ ; under this early-stopping protocol it drifts upward with γ , an effect Section 5 finds is not produced by checkpoint selection and is small relative to run-to-run variation.

γ	CI	CD	DLinear	CD/CI
0.0	1.0270	1.0287	1.0301	1.0016
0.3	1.0326	1.0393	1.0330	1.0065
0.6	1.0298	1.0377	1.0301	1.0077
0.9	1.0278	1.0362	1.0284	1.0082

Table 8 shows the CD/CI ratio across the (P, γ) grid. At $P \in \{4, 8, 16\}$ both modes run with full five-seed coverage and every entry is a genuine CD/CI ratio. At $P = 2$ CD remains infeasible under the non-fused attention implementation (Section 2). The cell therefore carries the CI-only test MSE where a run exists and a dash otherwise ($P = 2$ at $\gamma \in \{0.0, 0.3\}$ was not run under compute constraints). This joint sweep is a separate set of runs from the γ -sweep in Table 7, using a different data-generating protocol: iid innovations, $T = 20,000$, and a 70/10/20 split, against the γ -sweep’s compound-symmetry innovations at $\rho = 0.5$, $T = 14,400$, and 60/20/20 split (Table 1). Its $P = 16$ ratios sit nearer 1.001 than the 1.007–1.008 range reported there. Because no $P = 16$ cell was run under both generators, the protocol difference and per-seed variation cannot be separated with the runs available, and we make no attempt to attribute the gap to either. We therefore do not treat the two experiments as independent replications; what they jointly support is the weaker and sufficient claim that the CD/CI ratio at $P = 16$ stays within 1% of unity under both protocols.

Table 8: CD/CI MSE ratio in the (P, γ) boundary experiment. At $P \in \{4, 8, 16\}$ both modes run and the entry is a CD/CI ratio (above 1.0 favours CI). At $P = 2$ CD exceeds T4 memory under the non-fused attention implementation used here, and no ratio is defined; those cells carry the CI-only test MSE where a run exists and a dash where none was performed.

P	$\gamma = 0.0$	$\gamma = 0.3$	$\gamma = 0.6$	$\gamma = 0.9$
16	1.0012	1.0004	1.0005	1.0010
8	1.0014	1.0009	1.0009	1.0004
4	0.9997	1.0004	1.0002	1.0004
2	—	—	CI: 0.9806	CI: 0.9801



$P=4$ trained in a later software environment, CD at CI’s batch size; the remaining rows are the original environment. Ratios are within-cell and unaffected by both differences.

Figure 3: Leader-follower VAR(1): CD/CI test-MSE ratio across the (P, γ) grid, mean over seeds $\{42, 123, 456, 789, 1011\}$. Values above 1.0 favour CI. Coloured cells are the configurations where both modes were run ($P \in \{4, 8, 16\}$); the grey cells at $P = 2$ mark where CD exceeds T4 memory under the non-fused attention implementation and was not run. Eleven of the twelve coloured cells favour CI and one ($P = 4, \gamma = 0.0$) falls marginally below unity at 0.9997; the ratio stays within 0.15% of unity throughout.

4.4 Block-wise Attention Ablation

The results above use a CD encoder attending over all $C \cdot N$ tokens. A residual question is whether the null result depends on that unrestricted scope, or would hold even under an encoder that can only attend within the coupled leader-follower pairs. We implement a block-wise variant (CD_Block) that restricts attention to each leader-follower pair, treating the isolate channel as a singleton. This is the minimum scope under which a block-wise encoder can express the lag-1 coupling under study, and the implementation is parameter-matched to vanilla CD, so any difference traces to attention scope alone. We train it alongside same-environment CI and CD arms across the full γ sweep, five seeds each.

Table 9 reports the three-arm comparison. Across $\gamma > 0$ the block-wise variant sits between the two reference arms, with CD_Block/CI ranging 1.0045–1.0056 against CD/CI’s 1.0065–1.0082. The paired CD_Block–CI difference is not reliably distinguishable from zero at any γ (95% CIs span $[-0.0018, +0.0111]$ at $\gamma = 0.3$ through $[-0.0005, +0.0119]$ at $\gamma = 0.9$), while paired CD_Block–CD differences run negative at every γ , narrowing vanilla CD’s deficit, though none individually reaches significance. Restricting attention to within-pair blocks therefore narrows but does not close vanilla CD’s deficit against CI, and we report the block variant as intermediate rather than extending the practical-equivalence conclusion of Table 10 to it with the same certainty.

Table 9: Leader-follower VAR(1) at $P = 16$: block-wise cross-variate attention (CD_Block) alongside CI and CD. CD_Block restricts attention to leader-follower pairs (isolate as singleton), parameter-matched to vanilla CD. Mean test MSE over seeds $\{42, 123, 456, 789, 1011\}$.

γ	CI	CD	CD_Block	CD/CI	Blk/CI	Blk/CD
0.0	1.0270	1.0287	1.0273	1.0016	1.0003	0.9986
0.3	1.0326	1.0393	1.0372	1.0065	1.0045	0.9980
0.6	1.0298	1.0377	1.0353	1.0077	1.0054	0.9977
0.9	1.0278	1.0362	1.0335	1.0082	1.0056	0.9974

The same three-arm design extends to the AR(1) grid at $C = 84$, $\rho = 0.5$, where the flattened sequence reaches $C \cdot N = 5,292$ tokens and the grouping (28 contiguous blocks of 3) is necessarily neutral, since compound-symmetry covariance has no block structure to align with. These runs use a later training environment in which CD and CD_Block fit at batch 8 rather than the main grid’s batch 1, plus a fresh five-seed data generation at 14,400 timesteps; the cell is therefore self-contained and never paired row-by-row against Table 4. All three arms land together, with mean test MSE 1.0076 (CI), 1.0045 (CD), 1.0050 (CD_Block); paired CD–CI is -0.0031 (-0.31% of the CI mean, 95% CI $[-0.0105, +0.0042]$), CD_Block–CI is -0.0027 ($[-0.0093, +0.0039]$), and CD_Block–CD is $+0.0005$ ($[-0.0022, +0.0032]$); every interval spans zero. The point estimates all sit inside the 1% band; the CD–CI interval’s lower edge reaches -1.04% of the CI mean, marginally past the band on the CD-favourable side, which we state rather than round away. CD-family validation minima again fall early, at epochs 6–10 against CI’s 9–17.

4.5 Mechanistic Interpretation

The two leader-follower experiments differ in how sensitive the gap looks to γ . The boundary sweep gives a flat trend. Regressing its 40 paired CD–CI differences at $P \in \{8, 16\}$ on γ , with patch size as a covariate, gives a slope of -0.0006 MSE per unit γ (95% CI $[-0.0021, +0.0009]$, $p = 0.44$), and a seed-clustered version of the same regression gives an equally flat slope (95% CI $[-0.0029, +0.0018]$, $p = 0.62$), so the result does not hinge on treating seeds as independent. The dedicated $P = 16$ sweep, by contrast, shows a larger apparent trend under the original early-stopping protocol. The $P = 4$ cells are excluded from the slope because they were trained in a later software environment where CD ran at CI’s batch size rather than at batch 8, and pooling them would confound the coupling effect with the environment difference; their within-cell CD/CI ratios, unaffected by either difference, appear in Table 8. The environment difference itself is small and measured. Re-running the CI arm at $P = 4$ in the later environment moves the per- γ CI mean by $+0.0002$ ($\gamma = 0.6$) and $+0.0006$ ($\gamma = 0.9$), well inside the old-environment 95% half-widths of 0.0268 and 0.0301. The two sweeps agree that CD never holds a practically relevant lead; where they part ways is whether the gap drifts with γ .

Section 5 bears on this difference without settling it. The dedicated sweep’s trend is not produced by checkpoint selection (Section 5.1), and at $\gamma = 0.6$ the deficit is comparable in size to the run-to-run spread across reruns of that cell. The part of the gap that does track γ traces to the shared prediction head rather than the encoder (Section 5.3). These are two separate findings from two separate experiments, and neither substitutes for the other.

A representational reading of why CD gains nothing from the coupling is also available. At $P = 16$ and $S = 8$, position k of variate $j + 10$ sits beside position k of variate j in the $C \cdot N = 21 \times 63 = 1323$ -token sequence,

so a lag-1 cross-channel relationship maps to the same patch position in both sequences rather than to a phase offset between them. The lag-1 information is therefore already aggregated within that shared position, leaving cross-variate attention with no distinct inter-patch signal to route. CD and CI test MSEs stay within seed variance in every cell where both modes run ($P \in \{4, 8, 16\}$). The finest resolution, $P = 2$ with stride 1, where lag-1 coupling would fall across adjacent patches and a phase offset would be visible, is CI-only because CD exceeds T4 memory there under the non-fused attention implementation used for this paper’s runs. This account therefore cannot be tested directly and remains a conjecture rather than a result.

5 Fair-Protocol Robustness Controls

The leader-follower $P = 16$ sweep raises a question the main protocol cannot answer on its own. The apparent CD–CI gap grows with γ (Section 4), which could mean either CD is genuinely worse under stronger coupling, or the training protocol places CD at a disadvantage. This section tests both readings with four analyses (checkpoint selection, update budget, head design, and covariance structure) and then summarises the resulting effect sizes against a pre-specified threshold. Checkpoint selection, the first of these, does not explain the gap. We fix the practical-equivalence threshold at 1% of the CI mean test MSE, pre-specified before these analyses were run; differences smaller than this in either direction count as practically irrelevant regardless of statistical significance. This is an operational cutoff on observed effect sizes, not a formal statistical-equivalence claim over the reported confidence intervals. We apply no formal multiple-comparisons correction across the many cells and analyses reported here; the pre-specified 1% band, held fixed and applied uniformly rather than tuned to the data, is the safeguard we rely on against selective reporting, not a substitute for multiplicity-adjusted hypothesis testing. The regression slopes and subgroup contrasts reported outside the primary AR(1) factorial comparison are exploratory, describing the observed data rather than serving as confirmatory tests.

5.1 Overtraining and Selection

To test whether checkpoint selection explains the gap, we instrument one training run per (mode, seed) at the leader-follower $\gamma = 0.6$, $C = 21$ cell, logging validation MSE five times per epoch and training past the early-stopping horizon for all five seeds $\{42, 123, 456, 789, 1011\}$. Because mid-epoch validation runs in evaluation mode, it draws no random numbers and does not perturb the weight trajectory, so one trajectory supports every post-hoc selection rule we compare on it. Whether it reproduces the original run is a separate question, addressed next.

These are fresh instrumented reruns under the identical protocol (same seeds, steps per epoch 1005 for CD and 63 for CI), not a replay of the original trajectories. They land within one epoch of the original run’s best epoch in 6 of 10 runs and within two epochs in the rest. CD’s rerun test MSE comes in lower than its original on all five seeds (+0.0079 MSE, 95% CI [+0.0042, +0.0117]), while CI’s difference spans zero (−0.0019, 95% CI [−0.0084, +0.0046]). We cannot explain that CD-specific shift with the runs available, and we do not treat the diagnostic as reproducing the original numbers.

What the curves show is overtraining in CD. Across seeds, CD reaches its validation minimum at epoch 4 to 7 and then climbs, in seed-mean validation MSE, from 0.9865 at the minimum to 1.1999 by the end of training, while CI reaches its minimum later, at epoch 16 to 20, and stays near it (seed means 0.9863 to 1.0033). Figure 4 shows the asymmetry. The early-minimum shape recurs beyond this cell. In the $C = 84$, $\rho = 0.5$ three-arm run (Section 4.4) CD-family best epochs fall at 6–10 against CI’s 9–17, and the single-seed ECL comparison shows the same pattern at $C = 321$ (Section 3).

What the curves do *not* show is that overtraining produces the gap. The original protocol already restores the validation-best checkpoint, since in every committed row the recorded total step count equals best epoch times steps per epoch, meaning the reported test MSE is the minimum-validation checkpoint, never the checkpoint at which patience expired. Selecting each model at its validation minimum is therefore not a different rule, and on the diagnostic curves the two rules select the same checkpoint on every seed, to the last digit. The difference between the original +0.0079 MSE (95% CI [+0.0046, +0.0113]) and the diagnostic −0.0031 MSE (95% CI [−0.0109, +0.0046]) decomposes into three parts. Epoch-granularity selection contributes exactly

zero, the evaluation cadence contributes $+0.0012$ paired (per-mode epoch-boundary-minus-mid-epoch gaps: CD $+0.0005$, 95% CI $[-0.0049, +0.0059]$; CI -0.0007 , 95% CI $[-0.0062, +0.0047]$), and the original runs sit a further $+0.0098$ (95% CI $[+0.0001, +0.0195]$) above the reruns, a gap between two run sets rather than two selection rules.

We therefore withdraw the selection explanation for this cell. The deficit under the original protocol is $+0.77\%$ of the CI mean, inside the pre-specified band, and the run-to-run spread at five seeds is comparable to the effect itself, so we do not read the $\gamma = 0.6$ deficit as a stable architectural quantity. The same holds for every selection rule and protocol variant we tried at this cell, in one direction or the other, and Section 6 shows no cross-channel signal is available at this horizon to any predictor.

Candidate mechanisms for the early minimum. Selection artefacts do not produce the deficit, but the early-minimum-then-degradation shape itself is worth explaining. Three hypotheses are testable with cheap instrumentation: optimisation instability from CD’s longer flattened sequence, late-stage representational redundancy across variates, and cross-variate attention overfitting to spurious inter-channel structure. We log per-epoch pre-clip gradient norms for all three arms (CI, CD, CD_Block) and an encoder-representation participation ratio at the $\gamma = 0.6$ cell, collected during the block-attention sweep of Section 4.4. The gradient-norm evidence best matches the first hypothesis. CD’s final-epoch pre-clip gradient norms exceed the training clip threshold of 1.0 at every $\gamma > 0$ (1.17–1.23, against CI’s 0.87–0.90 and CD_Block’s lower still at 0.82–0.84), and of 216 nonfinite training steps recorded across the sweep, 115 occur in CD and 101 in CD_Block, none in CI. The participation-ratio panel does not support the second hypothesis as stated. CD’s encoder-output feature dimensionality rises over training (3.75 to 4.42; CD_Block more sharply, 3.87 to 8.92), while it is CI’s that declines (4.21 to 2.64), the opposite of what redundancy in CD would predict. Neither diagnostic speaks to the third hypothesis, which these instruments were never designed to detect. We read these diagnostics as consistent with the optimisation-instability account for vanilla CD specifically, not as proof of mechanism; they leave the behavioural finding above unchanged either way, since it does not depend on which mechanism turns out to be correct.

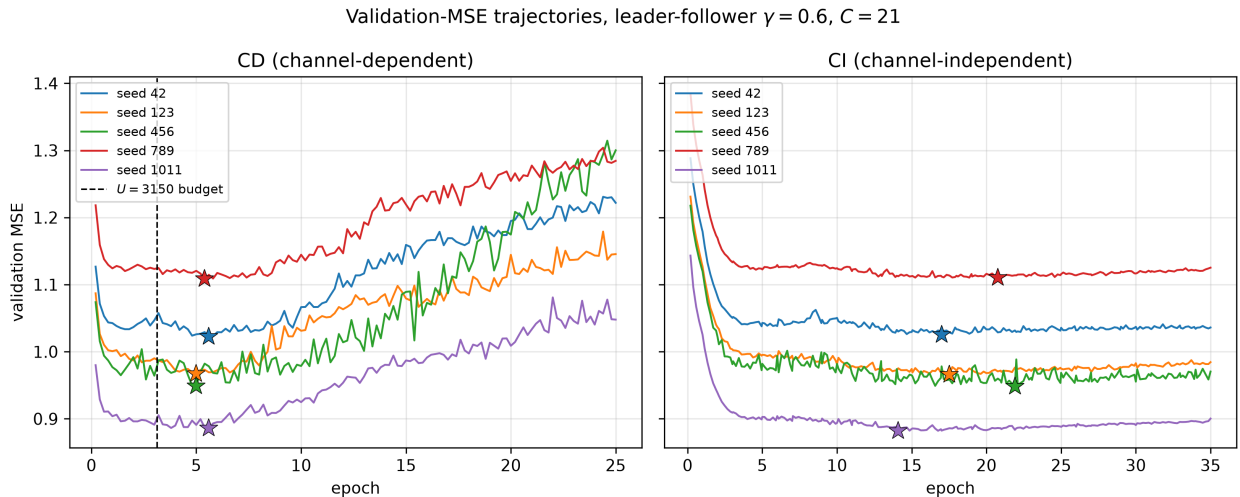


Figure 4: Validation-MSE trajectories at the leader-follower $\gamma = 0.6$, $C = 21$ cell, one instrumented rerun per seed, drawn to each mode’s diagnostic horizon. Stars mark each seed’s minimum on the mid-epoch validation grid. CD (left) reaches a sharp minimum near epoch 4 to 7 and then degrades, while CI (right) settles into a flat basin. The dashed line on the CD panel marks the matched update budget $U = 3150$, which falls before CD’s minimum, so a budget-capped rule truncates CD before its best checkpoint.

5.2 Matched Update Budget

At $C = 21$ the original protocol gives CD many more gradient updates per epoch than CI (Table 3), inviting the objection that CD is simply trained for longer and pushed past its optimum. Capping both modes at an identical update budget and taking the best checkpoint within it answers that objection directly, and the deficit does not survive the cap. The CD–CI difference is -0.0036 MSE (95% CI $[-0.0059, -0.0012]$) at leader-follower $\gamma = 0.0$ and -0.0053 MSE (95% CI $[-0.0092, -0.0014]$) at $\gamma = 0.6$, both favouring CD, and -0.0028 MSE (95% CI $[-0.0138, +0.0082]$) at the AR(1) $C = 84$, $\rho = 0.9$ cell, a tie, so the per-epoch step-count asymmetry does not produce the apparent deficit. We treat this protocol as a sensitivity check rather than a canonical schedule, because the budget $U = 3150$ is itself too tight for CD at leader-follower $\gamma = 0.6$, truncating training before CD’s validation minimum; under that cap the sign turns positive ($+0.0059$ MSE, 95% CI $[+0.0003, +0.0114]$). Given the overtraining shape documented in Section 5.1, that follows from where the cap falls rather than marking a separate disadvantage.

5.3 Head Fairness

The CD architecture in the main experiments shares CI’s per-variate prediction head, raising the residual objection that the head, not the encoder, limits CD. We rerun the leader-follower sweep with a cross-variate head (CD_Head) that pools across variates before projecting, leaving the encoder unchanged. The cross-variate head ties CI at every coupling strength, with CD_Head–CI at $+0.0016$ ($\gamma = 0.0$), $+0.0030$ ($\gamma = 0.6$), and $+0.0023$ MSE ($\gamma = 0.9$), all within 0.3% of the CI mean and all with 95% CIs spanning zero. Pooled across γ and clustered by seed, the CD_Head–CI grand mean is $+0.0023$ MSE ($+0.23\%$ of the CI mean, 95% CI $[-0.0023, +0.0070]$), so the cross-variate head is not detectably different from CI even pooled, well inside the 1% band. The plain shared-head CD over the same runs comes in at $+0.0017$, $+0.0079$, and $+0.0085$ MSE, carrying the γ -dependent widening the main sweep showed. Swapping only the head removes that widening while leaving the encoder fixed, consistent with the coupling-dependent widening originating in the shared head rather than in the cross-variate encoder. This is a head effect specifically, separate from the overtraining phenomenon of Section 5.1; neither explains the other.

5.4 Covariance Family

The AR(1) grid uses compound-symmetry covariance, where every channel pair shares the same correlation. To check that the null is not specific to that structure, we test a block-covariance family in which within-block correlation $\rho_{\text{in}} \in \{0.5, 0.9\}$ is concentrated in groups of seven variates and cross-block correlation is zero, giving CD a block structure to exploit that compound symmetry does not offer. CD gains no practically relevant advantage here either. At $C = 21$ the CD–CI difference is $+0.0028$ MSE ($+0.27\%$, 95% CI $[+0.0015, +0.0040]$) at $\rho_{\text{in}} = 0.5$ and -0.0062 MSE (-0.60% , 95% CI $[-0.0137, +0.0013]$) at $\rho_{\text{in}} = 0.9$; at $C = 84$, $\rho_{\text{in}} = 0.5$ it is $+0.0005$ MSE ($+0.05\%$, 95% CI $[-0.0005, +0.0016]$), and at $C = 84$, $\rho_{\text{in}} = 0.9$ it is $+0.0000$ MSE ($+0.00\%$, 95% CI $[-0.0010, +0.0010]$), a tie. Regressing the paired difference on variate count and ρ_{in} across the four cells gives a significant ρ_{in} slope (-0.0118 MSE per unit ρ_{in} , $p = 0.013$, $R^2 = 0.35$), so CD’s standing relative to CI improves slightly as within-block correlation rises, while variate count carries no effect ($p = 0.26$). Every cell sits within the 1% threshold, and the $C = 84$ cells again ran CD at batch size 1.

5.5 Practical-Equivalence Summary

Table 10 collects the controlled synthetic effect sizes with the selection rule used for each, so that cells evaluated under different protocols are never pooled. Across all tested AR(1), leader-follower, and block-covariance cells, the absolute CD–CI difference stays within the pre-specified 1% relative-MSE threshold in both directions. For the AR(1) grid this holds cell by cell as well as in aggregate. No single cell exceeds 0.56% of its CI mean (Section 3), so the grand-mean row of Table 10 never averages over any cell that breaches the threshold. Some differences prove statistically detectable in one direction or the other, but none reaches the practical threshold, so on these controlled synthetic processes neither mode holds a practically relevant accuracy advantage under fair selection. A related architecture-scope control, restricting CD’s attention to the coupled leader-follower pairs (Section 4.4), narrows vanilla CD’s deficit against CI but is not folded into Table 10. At five seeds it

is not confidently distinguishable from either reference arm, and we report it as intermediate rather than as a third practically-equivalent cell. Figure 5 makes the cost side of this trade-off directly visible. On two cells with substantial asymmetry in gradient updates to the selected checkpoint, CD attains comparable test MSE to CI while using several times more updates to get there. Each update is also more expensive under CD’s longer attention sequence, so the true compute gap is larger than the update-count ratio alone suggests. Table 3 reports updates per epoch; early stopping means each mode’s total updates to its own selected checkpoint, plotted here, is a different quantity, and the two ratios need not agree as a result. The leader-follower panel’s exact effect size is the $\gamma = 0.6$ early-stop row of Table 10; the AR(1) panel’s is given in Section 3.

Table 10: Controlled synthetic effect sizes, CD–CI in test MSE, each with the selection rule under which it was computed. Cells under different rules are never pooled. “Rel” is the difference as a percentage of the CI mean; the final column reports whether the absolute difference stays within the pre-specified 1% threshold. The $\gamma = 0.6$ early-stop row is from the original runs; the val-minimum row is from the instrumented reruns (Section 5.1), so the two differ in run set as well as rule.

Family	Cell	Selection rule	CD–CI	Rel	$\leq 1\%$
AR(1) grid	grand mean (9 cells)	early-stop	+0.0013	+0.13%	yes
Leader-follower	$\gamma = 0.6$	early-stop	+0.0079	+0.77%	yes
Leader-follower	$\gamma = 0.6$	val-minimum	−0.0031	−0.30%	yes
Leader-follower	$\gamma = 0.6$	matched- U	−0.0053	−0.52%	yes
Leader-follower	$\gamma = 0.9$	early-stop	+0.0085	+0.82%	yes
Leader-follower	$\gamma = 0.6$, cross-var. head	early-stop	+0.0030	+0.29%	yes
Leader-follower	$\gamma = 0.9$, cross-var. head	early-stop	+0.0023	+0.23%	yes
AR(1)	$C=84$, $\rho=0.9$	matched- U	−0.0028	−0.27%	yes
Block-cov.	$C=21$, $\rho_{in}=0.5$	early-stop	+0.0028	+0.27%	yes
Block-cov.	$C=21$, $\rho_{in}=0.9$	early-stop	−0.0062	−0.60%	yes
Block-cov.	$C=84$, $\rho_{in}=0.5$	early-stop	+0.0005	+0.05%	yes
Block-cov.	$C=84$, $\rho_{in}=0.9$	early-stop	+0.0000	+0.00%	yes

6 Theoretical Ceiling on Achievable Accuracy

The empirical near-parity result above raises a natural question: how much accuracy was actually available for CD to win. We answer this in closed form from the generative model itself, without additional training runs². For every diagonal-transition family (the AR(1) grid and block-covariance experiments), the argument is exact. Since innovations are drawn i.i.d. across time and the transition matrix carries no cross-channel term, the channel-independent predictor attains the same forecast-error variance as a full-information predictor, for any innovation covariance. In the paper’s z-scored units this reduces to $1 - \phi^{2h}$, independent of ρ , C , and group structure; at $\phi = 0.8$ and $\text{pred_len} = 96$ this equals 1.0 to float64 precision, indistinguishable from no signal.

The leader-follower design has a non-diagonal transition, and this argument therefore does not carry over directly to follower channels. Leader and isolate channels receive no incoming coupling and stay diagonal, so their ceiling is exact and matches the other families. For followers, both ceilings are available in closed form. The full-information (CD) bound follows directly from the joint state-space model. The own-history (CI) ceiling takes more care, because a follower’s past partially reveals its leader’s past, so the optimal own-history predictor is not the marginal AR(1) predictor. Treating the joint process as a state-space model observed through the follower channel alone, the steady-state Kalman predictor is optimal over all own-history predictors, and its h -step error variance is the CI ceiling. The derivation script asserts two consistency checks (to float64 tolerance) and raises on failure: at $\gamma = 0$ the ceiling must reproduce $1 - \phi^{2h}$ at every horizon, and $\text{CD} \leq \text{CI} \leq 1$ must hold at every cell.

²Reproducible from `derive_theoretical_bounds.py`, committed alongside the paper’s other analysis scripts; every value in Table 11 regenerates directly from this script against the same generator constants used elsewhere in the paper.

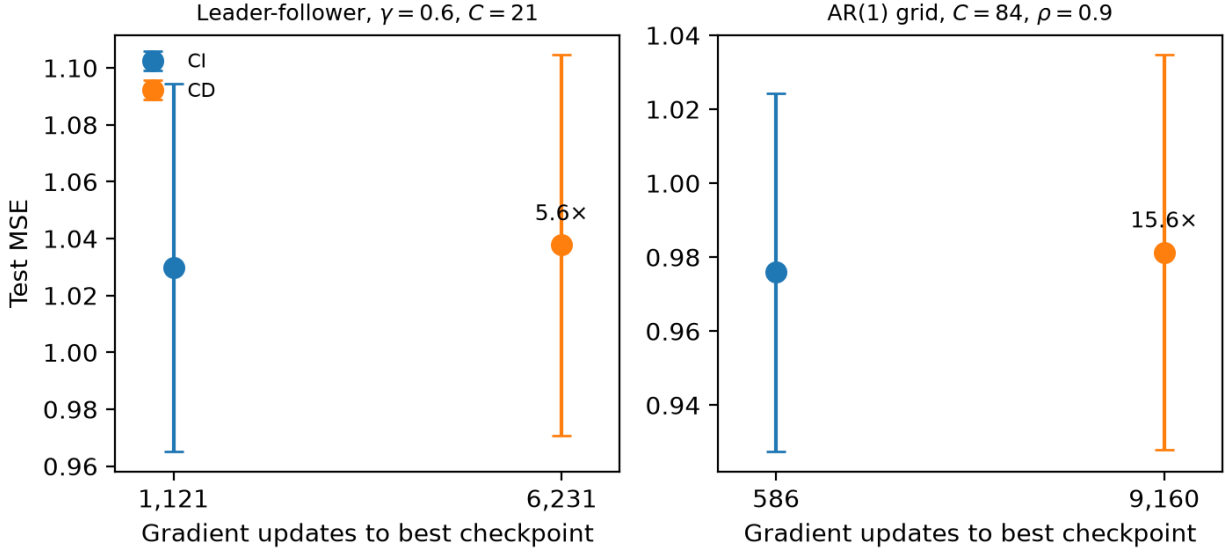


Figure 5: Test MSE versus total gradient updates to the validation-selected checkpoint (log scale), CI and CD, for two cells with substantial optimisation-effort asymmetry. This measures update count, not wall-clock compute, and each CD update is also more expensive, so these ratios understate the true compute gap. Points are means over five seeds; error bars show the cross-seed standard deviation. CD reaches essentially the same accuracy as CI while consuming $5.6\times$ (leader-follower, $\gamma = 0.6$) to $15.6\times$ (AR(1) grid, $C = 84$, $\rho = 0.9$) more gradient updates.

Table 11 reports both ceilings across the γ -by- h grid. At $h = 96$, the paper’s evaluation horizon, the full-information and own-history Bayes-optimal forecast-error ceilings coincide to float64 precision at every tested γ . The DGP therefore supplies no irreducible full-information accuracy advantage at this horizon; any empirical CD–CI difference observed there must arise from finite-sample, optimisation, or implementation effects rather than from a gap in the optimal forecast ceiling. At short horizons the ceilings separate, and the two natural framings of the opportunity peak at different points. The relative gap $(CI - CD)/CI$ is largest at $h = 1$ (9.8%, 24.9%, and 39.6% of the CI-achievable error at $\gamma = 0.3, 0.6, 0.9$), while the absolute gap peaks at $h = 5$ (0.059, 0.073, and 0.075 z-scored MSE; relative to the CI-achievable error at $h = 5$, nine to fifteen times the 1% criterion applied at $h = 96$). Whether CD converts that short-horizon opportunity into realised accuracy is untested here and is named in the Limitations; we make no claim about horizons this paper does not evaluate.

Table 11: Follower channel: closed-form ceilings on forecast MSE (z-scored units) across the leader-follower coupling sweep. CD is the full-information lower bound; CI is the ceiling for a predictor restricted to the follower’s own history (steady-state Kalman). At $h = 96$, the paper’s evaluation horizon, the two coincide to float64 precision at every tested γ .

h	$\gamma = 0.0$		$\gamma = 0.3$		$\gamma = 0.6$		$\gamma = 0.9$	
	CD	CI	CD	CI	CD	CI	CD	CI
1	0.3600	0.3600	0.1283	0.1423	0.0523	0.0696	0.0272	0.0450
5	0.8926	0.8926	0.5906	0.6493	0.4591	0.5320	0.4083	0.4831
10	0.9885	0.9885	0.8947	0.9199	0.8483	0.8796	0.8294	0.8615
50	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
96	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

7 Limitations

Every conclusion here is conditional on the generative families studied. The AR(1) process has a diagonal transition, Gaussian innovations, and compound-symmetry covariance; the leader-follower extension adds one specific coupling topology with a single scalar γ at $C = 21$. We do not explore leader-follower structures at higher variate counts (multiple leader-follower blocks embedded in $C = 84$, for instance). The covariance families tested, compound-symmetry on the AR(1) grid and block-diagonal in Section 5.4, do not exhaust the space; hierarchical or factor-structured correlations could behave differently even without lagged dependence, and neither is tested here. Richer coupling graphs, multi-order lag coupling, chaotic dynamical systems, non-Gaussian noise, and non-stationary dynamics fall outside scope as well. The AR(1) coefficient is fixed at $\phi = 0.8$ and all synthetic experiments use `pred_len = 96`; the conclusions are therefore specific to this dynamic regime and horizon. The horizon scoping matters. The 1% band was specified for $h = 96$ and cannot be carried to $h < 50$ without restating it against the ceilings of Table 11, which in turn fixes the target for the natural next experiment, a short-horizon comparison at $h \leq 10$, where the relative analytical CD–CI gap reaches 9.8–39.6% of the CI-achievable error at $h = 1$ against the 1% criterion used at $h = 96$.

The CD architecture used in the main experiments applies a per-variate head after the joint encoder, so cross-variate information from the encoder is never explicitly pooled during prediction. For RQ1 this is not a concern. The AR(1) design contains no lagged cross-channel signal by construction, so if the encoder attending across all $C \cdot N$ tokens produces no CD/CI gap, no head design could manufacture one from a signal that isn’t there. For the leader-follower experiments the head-fairness control in Section 5.3 addresses this objection in part. A cross-variate head ties CI at every γ and removes the coupling-dependent widening, consistent with that widening originating in the head rather than the encoder. What the control does not establish is that some other head design could turn the lagged signal into a genuine CD advantage; it shows only that the shared head, not an encoder bottleneck, drives the widening seen under the original protocol.

The $P = 2$ CD case was not evaluated. At $C = 21$ and $N = 511$, the CD sequence length of $21 \times 511 = 10,731$ tokens exceeds T4 VRAM at any batch size under the non-fused attention implementation used for this paper’s runs (the materialised-attention cost behind this bound is quantified for ECL in Section 2). Under fused attention kernels the memory wall itself is removed, since a probe at batch 64 fits CD at roughly 6.2 GiB peak allocation, but the cell becomes compute-bound: a timed-step projection puts a single CD seed at roughly 33 GPU-hours and the five-seed paired design at well over a hundred, so we report $P = 2$ CD as practically infeasible by wall clock rather than completing it. Only CI runs are available at $P = 2$, and only for $\gamma \in \{0.6, 0.9\}$; the remaining $P = 2$ cells were not run under compute constraints. This is the configuration where CD’s attention resolution most closely matches the lag-1 coupling scale, since one patch spans two timesteps and a lag-1 relationship falls across adjacent patches, leaving the presence or absence of a CD advantage there empirically untested. Patch resolution and prediction horizon are separate, complementary open dimensions. The short-horizon opportunity quantified above (Table 11) is a property of h alone, absent at this paper’s evaluation horizon regardless of P . Testing $P = 2$ together with a short-horizon evaluation, not $P = 2$ alone at $h = 96$, is what would determine whether finer patch resolution can convert the available short-horizon signal into measurable accuracy. The adjacent $P = 4$ configuration, initially excluded under the same assumed ceiling, was subsequently run in full and shows no CD advantage (Table 8), which narrows but does not close the untested region. A variate-token CD implementation, attending over C tokens rather than $C \cdot N$, would remove both the memory constraint and the quadratic-in- C compute cost, and would let the remaining $P = 2$ gap close. Building such a variant fell outside the scope of this work.

The $\gamma = 0.6$ diagnostic in Section 5.1 rests on one instrumented rerun per (mode, seed) at a single cell. Those reruns do not exactly reproduce the original runs. The CD arm’s rerun test MSE comes in lower than its original on all five seeds, and the cause is unidentified. Seeds, batch sizes, steps per epoch, and stopping rule are identical between the run sets and verified against the committed result rows, ruling out a configuration difference, and a consistent sign across five seeds is unlikely under nondeterminism alone. Nor is the exposure confined to one cell. Every CD figure in this paper comes from a single run of the same pipeline, with rerun evidence at this cell only, so we cannot bound the effect elsewhere. What we can bound is its consequence, a measured shift of +0.77% of the CI mean. Checking a uniform shift of this size against every synthetic cell individually, it stays inside the 1% band everywhere except one (block-covariance $C = 21$, $\rho_{\text{in}} = 0.9$,

measured -0.60% , which the shift would bring to -1.37%). Every other cell, meaning the full AR(1) grid, the leader-follower sweep, and the remaining three block-covariance cells, stays inside $\pm 1\%$ under the same hypothetical adjustment. Under the same correction the pooled ETTh1 deficit ($+0.0147$ MSE, $+2.6\%$ of the CI mean) falls to roughly $+1.8\%$, still above the band. We do not extend the overtraining measurement itself to other cells or datasets.

All architectural conclusions are confined to PatchTST with flattened-token channel dependence. Other ways of building channel dependence, such as hierarchical or graph-based encoders, may behave differently, and the results are in any case specific to the patch sizes and lookback windows used here.

8 Related Work

PatchTST (Nie et al., 2023) introduced patch-based channel-independent Transformers for long-horizon forecasting and showed competitive or superior performance against models that attend across variates; its own ablation already reports early overfitting in a channel-mixing variant relative to the channel-independent design (Nie et al., 2023, App. A.7.1). The benchmark results leave the mechanism open. The datasets confound patch size, weight sharing, and cross-variate correlation simultaneously, and the source of any CI advantage cannot therefore be isolated from the architecture alone. Two further analyses extend the picture: Han et al. (2024) formalise a capacity–robustness trade-off between the two strategies, showing CD’s higher representational capacity comes at the cost of robustness to distribution shift, while Chen et al. (2024) tie CD’s benefit to pairwise channel similarity, motivating a clustering strategy that interpolates between CI and CD. Neither makes the present controlled-DGP study redundant. Both characterise CI/CD behaviour on observational benchmarks where correlation strength, dimensionality, and coupling vary together, whereas this paper isolates each factor under a controlled generative model. Section 1 scopes the contribution accordingly, to that isolation rather than to a first general account of when CI/CD works.

DLinear (Zeng et al., 2023) established a surprisingly competitive baseline: a decomposition-linear model that splits each series into a moving-average trend and a residual, each mapped linearly from lookback to horizon. A Transformer that cannot beat it has not demonstrably learned useful temporal structure beyond what a linear filter captures.

iTransformer (Liu et al., 2024) inverts the embedding: each variate’s full history becomes a single token and attention operates across variates. It performs well on large multivariate benchmarks, but those benchmarks mix correlation strength, dimensionality, and temporal structure in ways that make attributing any gain to cross-variate attention specifically very difficult.

LIFT (Zhao & Shen, 2024) identifies leader-follower pairs empirically and prepends leader context to each variate’s input before a CI model, recovering cross-variate signal without full joint attention. It is the closest antecedent to the leader-follower design used here; its limitation in this context is that the datasets it uses are observational, meaning lag structure and instantaneous correlation cannot be decoupled.

Abdelmalak et al. (2025) show that standard benchmarks favour CI because they are dominated by within-channel periodicity, while datasets generated from chaotic ODEs with genuine dynamical coupling favour CD. The two papers are complementary rather than competing. Where Abdelmalak et al. contrast dataset categories, we fix the generative model analytically and sweep coupling strength continuously. The near-unity CD/CI ratios across all nine AR(1) cells (0.9984 – 1.0056) suggest that the ODE-dataset advantage of CD does not extend to the instantaneous-correlation regime, and the leader-follower experiment probes the boundary from the other side, finding the null extends there too, at least within the patch sizes and coupling strengths tested.

Read together, these five lines of work leave two gaps open. None runs a factorial experiment varying cross-variate correlation and dimensionality independently under a known generative model, so none can attribute performance differences to a single factor, and none analytically separates instantaneous correlation from dynamical coupling. LIFT probes lead-lag empirically on observational data, while Abdelmalak et al. build in strong ODE coupling but have no instantaneous-only baseline. The two synthetic experiments here address both gaps within a specific generative family, under the PatchTST architecture and training conditions described above.

9 Conclusion

We set out to find where in the coupling-structure space a channel-dependent PatchTST starts to beat its channel-independent counterpart. Within the ranges we tested it does not, and the fair-protocol controls point to cost, not accuracy, as the open question. On the AR(1) grid the grand-mean CD–CI difference is $+0.0013$ MSE with a pooled within-cell detection half-width of 0.0063 MSE over five seeds, and every cell difference falls inside that bound. Under leader-follower coupling a CD deficit appears to grow with γ on the original early-stopping protocol, though it stays inside the 1% band throughout. The diagnostic at $\gamma = 0.6$ documents CD overtraining (an earlier, sharper validation minimum followed by degradation), but the original protocol already restores validation-best weights, so overtraining cannot be what produces the gap. Fresh reruns of that cell, replaying the same early-stopping rule, give -0.0019 MSE (95% CI $[-0.0105, +0.0066]$) against the original $+0.0079$, a run-to-run spread comparable to the effect itself; selecting each rerun at its own validation minimum instead moves this slightly further in CD’s favour, to -0.0031 (Table 10). Matching the gradient-update budget rules out the per-epoch step-count asymmetry as one candidate cause. Swapping in a cross-variate head keeps CD_Head within 0.3% of CI at every γ and removes the γ -dependent widening, consistent with that widening originating in the shared head rather than the encoder, and a block-covariance family extends the no-advantage finding past compound-symmetry correlation. This null is not only empirical. The closed-form ceilings derived in Section 6 coincide at $h = 96$ for every tested coupling strength, so the near-parity result is exactly what the generative model predicts. No tested synthetic cell shows an absolute CD–CI difference outside a pre-specified 1% relative-MSE threshold in either direction, so on these controlled processes neither mode holds a practically relevant accuracy advantage once selected fairly.

The two modes diverge in cost. Making PatchTST channel-dependent by concatenating every variate’s patches into one attention sequence gives $C \cdot N$ tokens and an attention cost of $\mathcal{O}(C^2 N^2)$ against $\mathcal{O}(CN^2)$ for CI. That cost forces CD onto smaller batches and many more gradient steps per epoch, drops its batch size to one at $C = 84$ in the main grid, and at $C = 321$ (ECL) puts a CD epoch at roughly 956 s, about 4.5 GPU-hours for the completed $H = 336$ run, while CI trains comfortably inside the same budget. Variate-token designs such as iTransformer (Liu et al., 2024) attend over C tokens instead of $C \cdot N$ and sidestep this cost profile entirely; what we report is therefore specific to PatchTST’s flattened-token CD rather than to channel dependence in general. On a single-T4 budget CI is the better default on accuracy per unit compute. It matches or beats CD wherever both run, and CI remains trainable in the tested $P = 2$ cells where this CD formulation exceeds the available T4 memory under the non-fused attention implementation.

The observational results point the same way, though they sit outside the equivalence claim. On ETTh1 ($C = 7$) under a matched-budget protocol, CI holds a small but reliable edge, significantly better at two of four horizons, never worse, with a pooled CD–CI of $+0.0147$ MSE, about 2.6% of the CI mean. That figure sits above the 1% band we apply to the synthetic cells, so we report it as a genuine if modest CI advantage, kept separate from the equivalence result. If the run-set discrepancy documented at the $\gamma = 0.6$ cell held uniformly across CD runs, this figure would fall to roughly 1.8%, still above the band.

Several things stay open by design. The generative families are narrow (diagonal-transition AR(1), one leader-follower topology, and block-covariance, all at $\text{pred_len} = 96$ and, for leader-follower, at $C = 21$), and the short-horizon regime where Section 6’s ceilings actually separate goes untested here. The $P = 2$ CD configuration, where attention resolution most closely matches the lag-1 coupling scale, is the most direct comparison left undone. Its token sequence exceeds T4 memory under the non-fused attention implementation used here, and under fused attention the cell is bound by wall clock instead (Section 7); a variate-token CD implementation is the natural way to remove either constraint. The early-minimum-then-degradation shape CD shows under this protocol is now diagnosed, not explained: the evidence is consistent with optimisation instability, does not support representational redundancy as stated, and cannot speak to attention overfitting (Section 5.1). We do not treat that mechanism as settled. The natural next steps are testing whether the null holds at shorter horizons, at $P = 2$, and under a variate-token CD formulation.

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